Metadata for Scientific Data Integrity

for Energy

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Outline

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Target

Intro: Integrity to FAIR

Planning: The DMP

Operations: RDM in Practice

Metadata: Serialization

Metadata: Semantics

Publishing: KITOpen and OEP

Our Target: Publish a distribution of some datasets, on two different platforms

How we will Work

1. **W**e explore the core concepts and the "why"
2. **W**e look at practical cases:
   - Measured active power photovoltaic on the top roof of IAI, after aggregation
   - Data gathered from an MQTT Broker
   - The content of this training material for future reference

Outline

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Part 1: From Data Integrity to FAIR

09:15 - 09:45

Goal

To connect the foundational, ethical duty of **Data Integrity** to the modern, practical framework of the **FAIR Principles**.

Learning Objectives

- **Define** the core concepts of **Data Integrity** (accuracy, completeness, and consistency) and the four pillars of the **FAIR** acronym.
- **List** three Guidelines from the DFG code of conduct concerning Data Integrity.

Theory: What is Data Integrity?

Define

Data Integrity is the maintenance and assurance of the **accuracy**, **completeness**, and **consistency** of data over its entire life-cycle.

- **Accuracy:** Data reflects the real-world event. It is correct and precise.
- **Completeness:** No critical data is missing. The dataset is whole.
- **Consistency:** Data is uniform. The same value is represented the same way everywhere (e.g., 'KIT' vs. 'Karlsruhe Institute of Technology').

Theory: What are the FAIR Principles?

Define

The FAIR Principles are a set of guiding principles to make research data more valuable and reusable.

- **F**indable: Data (and metadata) must be easy to find by both humans and machines. (e.g., Persistent **I**dentifiers - PIDs).
- **A**ccessible: Once found, the (meta)data must be retrievable via a standard protocol. (e.g., HTTP, FTP).
- **I**nteroperable: Data must be able to be combined with other datasets, using formal, shared languages and vocabularies.
- **R**eusable: Data must be richly described with metadata and have a clear license so it can be replicated or combined.

More details? GOFAIR

Concept: DFG Code of Conduct

List

Data Integrity is a core component of the DFG "Guidelines for Safeguarding Good Research Practice." Deutsche Forschungsgemeinschaft, 2025

Key Guidelines on Integrity

- **Guideline 7 (Safeguarding and Storing):** Researchers safeguard their research data and document the steps taken.
- **Guideline 12 (Documentation):** Documentation and re- search results must not be manipulated; they are protected as effectively as possible against manipulation.
- **Guideline 17 (Archiving):** Primary data must be securely archived for a minimum of ten years.

Interactive: Defining Our Terms

Practice

Group Discussion (8 min)

Let's put this together.

- In your own words, what is the biggest threat to **Data Integrity** in your specific research field?
- Which of the four **FAIR** principles do you find the most challenging to implement?
- How do **Guideline 12 (Documentation)** and the 'R' (Reusable) principle relate?

∴ Let's miro (How do GRP12 and 'R' relate)?

Learning Objectives

- **Explain** the **causal relationship** between maintaining high **Data Integrity** during collection and achieving the **Reusability** ('R') principle.
- **Explain** how adhering to the FAIR-Principles is crucial to reach accordance with the DFG guidelines.

Theory: The Integrity-Reusability Chain

Explain

High Data Integrity → Trust → Reusability

- **Reusability ('R')** is the ultimate goal of FAIR. It means someone else (or you, in 5 years) can understand, trust, and reuse your data.
- This is **impossible** without Data Integrity.
- If your data is **inaccurate**, reusing it gives wrong results.
- If your data is **incomplete**, reusing it is impossible.
- If your data is **inconsistent**, reusing it requires massive, error-prone cleaning.

Concept: The Broken Chain

Explain

Example of Low Integrity

A research group publishes a dataset on material strength.

- Column ‘temp‘: Contains values ‘25C‘, ‘298K‘, and ‘25‘.
- Column ‘pressure‘: Contains values ‘1 atm‘, ‘101.3 kPa‘, and ‘1.013‘.

This data is **inconsistent**.

Impact on Reusability

A second researcher tries to reuse this data for a meta-analysis. They cannot trust the values. They cannot automate a script. The data is **not reusable**. The ’R’ principle fails.

Theory: FAIR as a Tool for DFG Concordance

Explain

The DFG Guidelines tell you **what** to do (e.g., "Archive data", "Document results").
The FAIR Principles tell you **how** to do it effectively for the modern, digital age.

- **DFG Guideline 7 (Cross-phase quality assurance)** → Is achieved by implementing 'R' (Reusable), which requires rich metadata.
- **DFG Guideline 12 (Documentation)** → Is achieved by implementing domain specific guidelines for assessment and review, along with strong provenance, enabling Interoperability (I).
- **DFG Guideline 13 (Providing public access to research results)** → Is achieved by implementing 'F' (Findable) with a PID and 'A' (Accessible) in a repository.

∴ Let's miro (Categories of Metadata for DFG Compliance)?

Interactive: The Broken Chain

Practice

Think-Pair-Share (10 min)

Scenario: A researcher leaves KIT. Their raw data is on a USB drive in their desk. The only "documentation" is their published paper. **Discuss:**

- Which **Data Integrity** principles are broken?
- Which **FAIR** principles are broken?
- Which **DFG Guidelines** are broken?

Goal: **Explain** how these three concepts are deeply linked.

Learning Objectives

- **Apply** data validation and cleaning techniques (e.g., scripting checks, normalisation) to a raw dataset.
- **Write** a checklist to help Researchers assess their own concordance with the guidelines.

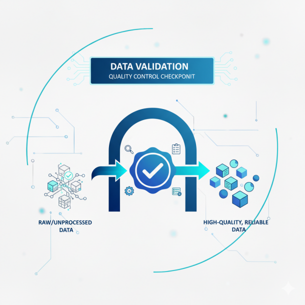
Theory: What is Data Validation?

Apply

Data Validation is the set of processes used to ensure data has high **integrity** (accuracy, completeness, consistency).

Techniques:

- **Range Checks:** Is a pH value 15? (Invalid)
- **Format Checks:** Is a date 'YYYY-MM-DD'?
- **Normalisation:** Converting all units (C, K) to one standard (K).
- **Schema Validation:** Does the data conform to the rules we set?



Validation as a Quality Control

Concept: Validation in Practice (Pseudo-Code)

Apply

You don't have to check data manually! Use scripts.

Simple Python Check (Pseudo-code)

```
data = read_csv("my_experiment.csv")
for row in data:
    # Check for completeness (e.g., no missing IDs)
    if row['sample_id'] is None: print("Error: Missing sample_id!")
    # Check for consistency (e.g., standardising units to Kelvin)
    if row['unit'] == "C": row['temp_K'] = convert_to_kelvin(row['temp'])
    # Check for range (e.g., no impossible values below absolute zero)
    if row['temp_K'] < 0: print("Error: Impossible temperature!")
save_csv("my_clean_data.csv", data)
```

Interactive: The Concordance Checklist

Practice

Activity: Write a Checklist (15 min)

You are onboarding a new PhD student. **Task:** Draft a 5-point "Data Integrity Checklist" to help them assess their own concordance with DFG guidelines.

- *Example 1: (Completeness)* "Does every dataset have a 'README' file?"
- *Example 2: (Consistency)* "Is there a script to check that all units are standardised?"
- *Example 3: (Accuracy)* "Has a colleague double-checked the manual data entry?"

Goal: Write this checklist. We will share them.

∴ Let's miro (Onboarding)

Learning Objectives

- **Differentiate** between the requirements for data **Authenticity** (Integrity) and data **Accessibility** (FAIR).
- **Identify** criteria of publication platforms.

Theory: Authenticity vs. Accessibility

Differentiate

These two concepts are often confused, but are very different.

Authenticity (Data Integrity)

Question: Is this the *real, unaltered* data?

Method:

- Checksums (e.g., MD5, SHA256)
- Digital Signatures
- Audit Trails
- Version Control History

Metaphor: A sealed, tamper-proof envelope.

Accessibility (FAIR Principle 'A')

Question: Can I *get* to the data?

Method:

- A Persistent Identifier (PID)
- A standard protocol (e.g., HTTP)
- An open, public URL
- Clear authentication (if not open)

Metaphor: A public mailbox with a clear address.

Concept: PIDs as the Bridge

Differentiate

Persistent Identifiers (like a DOI) are the bridge between Authenticity and Accessibility.

- A PID (like a DOI from zenodo) makes the data **Accessible** ('A') by giving it a permanent, findable web address.
- The repository *behind* the DOI (zenodo) ensures **Authenticity** (Integrity) by storing the file and often providing a checksum (MD5/SHA) to prove it hasn't been altered since publication.

Key Idea

You can have Authenticity without Accessibility (e.g., a file on a secure, offline server). You should *never* have Accessibility without Authenticity (e.g., a public file that anyone can edit).

Theory: Criteria for Publication Platforms

Identify

When choosing *where* to publish, you must **identify** its criteria.

- **Findability-Degree:** This is key.
- **Level 1 (Low):** Publishes data (e.g., a project website).
- **Level 2 (Good):** Publishes data + assigns a PID (e.g., KITOpen).
- **Level 3 (Excellent):** Publishes data + PID + rich, machine-readable, *domain-specific* metadata (e.g., Open Energy Platform).

Other Criteria

Check for: PID minting (DOIs?), license support (CC-BY?), long-term preservation plan, costs, access control (embargoes?).

Interactive: Platform Criteria

Practice

Discussion (10 min)

Task: Let's **identify** criteria for publication platforms.

- What is the "findability-degree" of putting your data on a personal website?
- What about on a GitHub repository?
- What about on zenodo?

Question: Why is a high "findability-degree" so important for both you (impact) and your field (progress)?

More? Koubaa, 2025

Learning Objectives

- **Appraise** a current data storage system against both the criteria for long-term **Data Security** (Integrity) and **machine accessibility** ('A' of FAIR).
- **Describe** the importance of the find-ability-degree in selecting where to publish.

Theory: Criteria for Storage vs. Access

Appraise

We must evaluate storage on two axes: Safety (Integrity) and Openness (FAIR).

Data Security (Integrity)

Question: Is my data safe?

- **Durability:** Is it backed up?
- **Security:** Who has access?
- **Auditing:** Can I see who changed what? (Version history)

Machine Accessibility ('A')

Question: Can a machine get it?

- **Protocol:** Is it on 'http' or 'ftp'?
- **Identifier:** Does it have a PID/DOI?
- **Metadata Access:** Is metadata accessible, even if data is restricted?

Concept: Evaluating Storage Options

Appraise Storage Solutions

Let's appraise common storage systems against our ethical duties.

Storage System	**Data Security (Integrity)**	**Machine Access ('A')**
Your Laptop HD	**Very Low** (No backup, easily lost)	**None** (Offline)
Shared Drive	**Medium** (Backed up, access control)	**Low** (Internal, not public)
GitHub (Public)	**High** (Versioned, backed up)	**Medium** (No PID, not a formal repo.)
KITOpen (Repo.)	**Very High** (Managed, backed up)	**Very High** (Public, HTTP, PID)

Interactive: Appraise Your Storage

Practice

Personal Evaluation (10 min)

Task: Think about the *primary* place you store your "in-progress" research data right now.

- **Appraise** its **Data Security**. What is your biggest risk? (e.g., "My laptop gets stolen," "A colleague overwrites the file.")
- **Appraise** its **Machine Accessibility**. Could an external collaborator (or a script) access it without you emailing it?

Share: Let's hear some examples.

Learning Objectives

- **Develop** a comprehensive, integrity-focused **Metadata Standard** that, when implemented, ensures the data meets all FAIR principles.
- **Distinguish** different types of interoperability (The 5 Stars).

Theory: What is a Metadata Standard?

Develop

A Metadata Standard is not just a list of fields. It's a complete **"instruction manual"** for your data, essential for maintaining data integrity (an example?).
It must define:

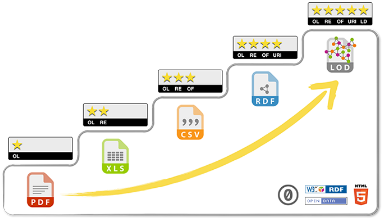
1. **Schema:** What fields are required? (e.g., `Creator`, `Date`, `Sample_ID`).
2. **Vocabularies:** What are the allowed *values* for a field? (e.g., For `Unit`, you must use `K`, `m`, or `kg`. No `Celsius` or `grams`.)
3. **Format:** How is it written? (e.g., "All metadata will be stored as a `metadata.json` file alongside the data.")

Key Idea

An "integrity-focused" standard **requires** fields that ensure accuracy (e.g., `Instrument_Calibration_Date`) and consistency (e.g., `Unit_Standard`).

Concept: The 5 Stars of Interoperability

Distinguish



Berners-Lee, 2010

- ⋆ – Data is on the web (e.g., a PDF)
- ⋆⋆ – Machine-readable (e.g., an Excel file)
- ⋆⋆⋆ – Non-proprietary format (e.g., a CSV file)
- ⋆⋆⋆⋆ – Uses URIs/PIDs as identifiers
- ⋆⋆⋆⋆⋆ – **Links** to other data (Linked Open Data)

This is the ∴ ladder we are trying to climb!

Interactive: Brainstorm Your Standard

Practice

Workshop (8 min)

Task: Let's **develop** a *minimal* metadata standard for a simple experiment. **Scenario:** Measuring the temperature of 10 water samples. **Brainstorm:** What fields are *essential* for...

- **Integrity:** (e.g., `operator_name`, `calibration_date`)
- **FAIRness:** (e.g., `sample_id`, `unit`, `location_geocoordinates`)

Goal: Distinguish between fields for simple documentation and fields for 5-star interoperability.

∴ Let's miro (temperature of 10 water samples)

Summary

- **Data Integrity** (Accurate, Complete, Consistent) is the foundation.
- **FAIR Principles** are the practical framework for modern data.
- **DFG Guidelines** are the "law" that requires both.
- We apply integrity by **validation** and **appraising** our tools.
- We plan for FAIR by developing **Metadata Standards**.
- RDA Guidelines for publishing structured metadata on the web Wu et al., 2021 can be considered.

End of Part 1

Bibliography

Berners-Lee, T. (2010). 5 ⋆ OPEN DATA: The 5-Star Deployment Scheme for Linked Open Data [Accessed on 2025-11-18]. https://5stardata.info/en/

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Koubaa, M. A. (2025, October). Energy Platforms Survey. https://doi.org/10.5281/ZENODO.17340933

Wu, M., Juty, N., WG, R. R. M. S., Collins, J., Duerr, R., Ridsdale, C., Shepherd, A., Verhey, C., & Castro, L. J. (2021, October). Guidelines for publishing structured metadata on the web. https://doi.org/10.15497/RDA00066

Outline

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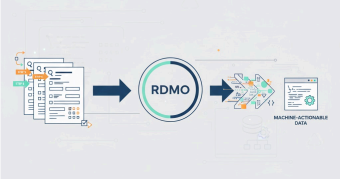
Day 1 - Part 2: The Planning Phase

10:50 - 11:35

Module 2 Goal

To master the **Data Management Plan (DMP)** as our primary tool for *planning* a FAIR publication, using the **RDMO** tool.



A DMP is the blueprint for your research data.

Learning Objectives (2.1)

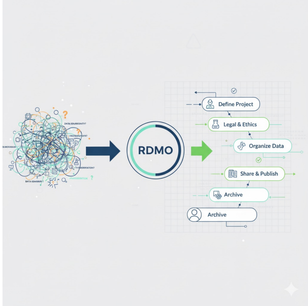
- **Define** the four core elements of the FAIR acronym (Refresher).
- **Identify** the specific RDMO question categories (e.g. *Metadata*, *Documentation*, *Long-term Preservation*) that correspond directly to each of the four FAIR principles.

Theory: What is RDMO?

Module 2.1: Identify

The **Research Data Management Organiser (RDMO)** is a tool to support the systematic planning, implementation, and management of research data.

- It's a "smart questionnaire."
- It guides you through the questions funders (like DFG) want answered.
- It generates a human-readable and machine-readable DMP.
- It is hosted by KIT!



RDMO turns chaos into a structure

Concept: Mapping FAIR to RDMO

Module 2.1: Identify

RDMO is structured around the RDM lifecycle, which maps directly to FAIR.

- **To be 'Findable' (F)...**
- *RDMO asks about:* "Project & Data," "PIDs," "Repository."
- **To be 'Accessible' (A)...**
- *RDMO asks about:* "Access & Re-use," "Legal & Ethical."
- **To be 'Interoperable' (I)...**
- *RDMO asks about:* "Metadata," "Vocabularies," "Formats."
- **To be 'Reusable' (R)...**
- *RDMO asks about:* "Documentation," "Licensing," "Quality."

Interactive: RDMO Treasure Hunt

Module 2.1: Practice

Activity: Log into RDMO (10 min)

Task: Log into the KIT RDMO instance. Create a new dummy project (use the "DFG" template).

Challenge: Identify the *exact question* in RDMO that asks...

- How you will license your data ('R')?
- What metadata standard will you use ('I')?
- If your data will have a PID ('F')?

Goal: Identify how RDMO is structured around FAIR.

URL: https://rdmorganiser.github.io & rdmo-docker-compose

Learning Objectives (2.2)

- **Explain** *why* addressing the 'F' (Findable) principle is critical during the *planning* phase.
- **Explain** how the structured output format of the RDMO tool supports the **Interoperability** ('I') of the DMP itself.

Theory: Why 'Findable' is a Planning Task

Module 2.2: Explain

You can't make data "findable" *after* the project is over. It's too late.

- **Planning for 'F'** means deciding *before you collect data*:
- **Where** will it be published? (e.g., KITOpen)
- **What** is the PID strategy? (e.g., We will get a DOI)
- **How** will it be described? (e.g., We will use Dublin Core)

If you don't plan for this, you end up with data on a hard drive that nobody (not even you) can find.

Draw the map *before* you start the journey

Theory: RDMO for Interoperability ('I')

Module 2.2: Explain

The DMP from RDMO isn't just a text document. It's a structured, machine-readable database entry.

The 'I' of the DMP Itself

You can **export** your DMP from RDMO in formats like:

- XML
- JSON
- etc. (RDMO-Instance dependant)
- This means your *plan* is now interoperable. It can be automatically read by funders (DFG), repositories (KITOpen), or other project management tools.
- This is "FAIR for your DMP."

Interactive: The "Why" of Planning

Module 2.2: Practice

Group Discussion (10 min)

Scenario: A researcher says, "I'm too busy doing research to write a DMP. I'll document my data when I'm finished." **Task:**

- **Explain** to this researcher why this is a bad idea, using the 'F' (Findable) principle.
- **Explain** how spending 2 hours on RDMO now will save them 20 hours later.

Learning Objectives (2.3)

- **Determine** the appropriate persistent identifier (e.g. DOI, Handle, ARK) to be used for the planned dataset.
- **Determine** and input the required **Persistent Identifier (PID) strategy** into the RDMO questionnaire.

Theory: The PID Landscape

Module 2.3: Determine

A Persistent Identifier (PID) is a long-lasting reference to a digital object.

Common PIDs

- **DOI (Digital Object Identifier):** Most common for publications and datasets. Excellent for citation.
- **Handle:** The system that DOIs are built on.
- **ARK (Archival Resource Key):** Common in libraries and archives.
- **ORCID:** A PID for *people*.

Your Strategy

For 99% of researchers at Helmholtz, the strategy is simple: "I will deposit my data in a repository (like KITOpen) that will mint a **DOI** for me." Your job is not to *create* the DOI, but to *plan* on getting one.

Concept: Why Not Just Use a URL?

Module 2.3: Determine

- A standard URL ('http://my-lab.kit.edu/data.zip') is **not persistent**.
- What happens when the website is redesigned? The server is changed? The researcher leaves?
- → **Link Rot!** The link is broken. The data is lost.
- A **PID** (like a DOI) points to a *resolver* (e.g., doi.org), which then redirects to the *current* URL. If the URL changes (e.g., KITOpen moves servers), the repository updates the resolver. The DOI never changes.

Interactive: Your PID Strategy

Module 2.3: Practice

Workshop: Fill in the DMP (10 min)

Task: Go back to your draft RDMO plan.

1. Find the section on "PIDs" or "Identifiers."
2. **Determine** your PID strategy.
3. **Input** your strategy.

Example Answer: "We will publish the final dataset in KITOpen to obtain a DOI. We will use this DOI in all publications that cite the data. Internal datasets will be identified using the HMC naming convention."

Learning Objectives (2.4)

- **Distinguish** between an *open* license (CC-BY) and a *restrictive* license (CC-BY-NC-ND).
- **Justify** the selection of a specific **data access restriction model** (e.g., under embargo) in RDMO.

Theory: The License Spectrum

Module 2.4: Distinguish

A license is the legal tool that makes data **Reusable ('R')**.

Creative Commons (CC) Licenses ∴.

- **CC0 (Public Domain):** No rights reserved. The best for data!
- **CC-BY (Attribution):** Anyone can use it, as long as they cite you. **(Recommended)**
- **CC-BY-SA (ShareAlike):** They must also share their new work under CC-BY-SA.
- **CC-BY-NC (Non-Commercial):** Cannot be used for commercial purposes. (Restricts 'R'!)
- **CC-BY-ND (No-Derivatives):** Cannot be changed. (e.g., can't combine it with other data. Kills 'R'!)

Theory: Access Model vs. License

Module 2.4: Justify

These are two different things!

- **License:** What can people *legally* do with your data *if* they have it? (e.g., CC-BY)
- **Access Model:** *Who* can get the data in the first place?
  - **Open Access:** Everyone.
  - **Embargoed:** Open, but only *after* a certain date (e.g., 1 year after publication).
  - **Restricted:** Only specific people (e.g., collaborators) can access it after application.
  - **Closed:** No one can access it.

Key Idea

You can have an **Open License** (CC-BY) on **Restricted Data**. This means "If I approve your access, you are then free to use this data as long as you cite me."

Interactive: Debate - Access vs. Restriction

Module 2.4: Practice

Group Debate (15 min)

Scenario: Your data is publicly funded (DFG) but contains potentially patentable information.

Group 1: Argue for...

- Open Access
- CC-BY License

(Hint: DFG requirements, fast science)

Group 2: Argue for...

- Embargoed Access
- Restrictive License

(Hint: KIT tech transfer, patents)

Goal: Justify your data access restriction model.

Learning Objectives (2.5)

- **Justify** the selection of a discipline-specific metadata standard over a general standard.
- **Appraise** a draft DMP against the DFG "Checklist for Handling of Research Data."

Concept: The DFG Checklist

Module 2.5: Appraise

The DFG provides a "Checklist for Handling of Research Data" in its proposals. This is the list the reviewers will use to **appraise** your plan.

Key DFG Checklist Items

- "What data will be collected?"
- "What metadata standards will be used?"
- "Who is responsible for the data?"
- "How will the data be archived for 10 years?"
- "What costs are associated with this?"

Interactive: Peer Review Your DMP

Module 2.5: Practice

Peer Review (15 min)

Task: Swap your draft RDMO plan (or the link) with a partner. **Review: Appraise** your partner's DMP using the DFG Checklist.

- Can you find the answer to "What metadata standard will be used?"
- Is their plan for 'Reusability' (licenses, documentation) clear?
- Give one piece of constructive feedback.

Goal: Appraise a draft DMP and help improve it.

Learning Objectives (2.6)

- **Develop** a comprehensive DMP section for the 'A' (Accessible) principle.
- **Generate** a complete, funder-compliant DMP for a research proposal using the RDMO tool.

Theory: Sub-Principles of 'Accessible'

Module 2.6: Develop

'Accessible' is more than just "it's on the web."

- **A1:** (Meta)data are retrievable by their identifier using a standardised communications protocol.
- *Your DMP Entry:* "Data will be deposited in KITOpen, which provides retrieval via HTTP and assigns a DOI."
- **A1.1:** The protocol is open, free, and universally implementable.
- *Your DMP Entry:* "The chosen protocol is HTTP(S), which is open."
- **A1.2:** The protocol allows for an authentication and authorisation procedure, where necessary.
- *Your DMP Entry:* "KITOpen manages authorisation."
- **A2:** Metadata are accessible, even when the data are no longer available.
- *Your DMP Entry:* "The repository (KITOpen) guarantees metadata persistence even if the data is removed."

Interactive: Finalize Your DMP

Module 2.6: Practice

Workshop (20 min)

Task: Go back to your RDMO plan one last time.

1. Review your answers for 'Access' (Module 2.4) and 'Metadata' (Module 2.5).
2. Ensure every question has a clear, concise, and specific answer. (e.g., change "a repository" to "KITOpen").
3. Click the "Export" button and look at the generated document.

Goal: Generate a complete, funder-compliant DMP.

Module 2: Summary

- The **DMP** is the central *planning* document for FAIR.
- **RDMO** is the tool that guides us through this plan.
- We must plan for **PIDs ('F')**, **Licenses ('R')**, **Access ('A')**, and **Standards ('I')** from Day 1.
- A good DMP is the **blueprint** for your research and the key to your funding proposal.

End of Part 2

Outline

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Day 1 - Part 3: RDM in Research Operations

Module 3 Goal

A plan is useless without action. We will learn to integrate RDM (ELNs, Git, Pipelines) into our **daily research operations**.

Learning Objectives (3.1)

- **List** the mandatory data records (e.g., metadata, raw data, scripts) required for project closure.
- **Identify** the mandatory metadata fields (e.g., sample ID, instrument ID) to be logged in an **ELN** upon acquisition.

Theory: The "Project Closure" Package

Module 3.1: List

When your project ends, you must archive a "research package" that ensures Reusability and meets DFG's 10-year rule.

Mandatory Records for Archiving

- **1. Raw Data:** The original, unaltered files from the instrument.
- **2. Processed Data:** The final, clean data used for figures.
- **3. Metadata:** The 'README' or 'metadata.xml' that describes everything.
- **4. Scripts:** The code (e.g., Python, R, Matlab) used to get from Raw to Processed.
- **5. Documentation:** The ELN export, codebook, or lab journal.

Theory: The ELN at Acquisition

Module 3.1: Identify

To build that archive, you must capture metadata *at the moment of acquisition* in your **Electronic Lab Notebook (ELN)**.

Mandatory ELN Fields
zb_med-informationszentrum_lebenswissenschaften_electronic_2021

Your ELN template for a new experiment should **force** you to enter:

- ‘`experiment_ID`‘, ‘`operator_name`‘
- ‘`date_time`‘ (timestamp), ‘`instrument_ID`‘
- ‘`sample_ID(s)`‘, ‘`project_name`‘
- ‘`hypothesis`‘ (Why are you doing this?)

If you don't record this *now*, it is lost forever. This is the core of operational data integrity.

Interactive: Design Your ELN Template

Module 3.1: Practice

Activity: Your ELN Template (10 min)

Task: You are setting up a new ELN (e.g., https://demo-kadi4mat.iam.kit.edu). You are creating a template for a "standard experiment" in your field. **Brainstorm:**

- **List** 5-10 **mandatory metadata fields** you must capture.
- **Identify** the *one* field that is most often forgotten, but causes the most problems later.

Goal: Identify mandatory fields for your ELN. .·.

Learning Objectives (3.2)

- **Summarise** the differences in data access controls and their impact on operational security.
- **Illustrate** how connecting an **Instrument Registry** to the ELN automatically fulfils 'F' and 'I'.

Theory: Access Controls in Operation

Module 3.2: Summarise

Access controls are not just for publishing. They are for *operational security* during the project.

- **Problem:** Your "in-progress" data is often your most sensitive asset.
- **Risk 1 (Too Open):** Data on a shared drive with no access control. A student from another group accidentally deletes it. (Integrity failure!)
- **Risk 2 (Too Closed):** Data on your personal laptop. If you get sick, your team cannot continue working. (Operational failure!)
- **Solution:** Use **role-based access**.
  - 'Project_Leader': Read/Write
  - 'PhD_Student': Read/Write
  - 'Master_Student': Read-Only (on raw data)
  - 'External_Collaborator': Read-Only (on processed data)

Theory: The Instrument Registry

Module 3.2: Illustrate

Manual Way (Low 'I')

ELN Field: ‘instrument_name‘
You type: ”The new microscope in room 205”
Problem: This is free-text. It’s not machine-readable. It’s not interoperable.

Automated Way (High 'I')

An **Instrument Registry** is a database of all lab equipment.
ELN Field: ‘instrument_ID‘
You select: ‘HMC-KIT-Microscope-047‘
Result: Your data is now *linked* to that instrument’s entire history (calibration, model, etc.).
This is **Interoperability (’I’)** and **Findability (’F’)**!

Interactive: Your Security Workflow

Module 3.2: Practice

Discussion (10 min)

Task: Summarise the access controls on your current project's "in-progress" data.

- Who has access?
- Is it too open or too closed?
- How does this impact your operational security?

Illustration:

- **Illustrate** (by drawing a diagram) how connecting your lab's instrument list (even a simple Excel file) to your ELN would improve your data's 'I' and 'R' value.

ex. Sensor Management System delivers PIDs like :
https://hdl.handle.net/21.11157/b88fe935-2072-4106-8aca-f008e2feab48

Learning Objectives (3.3)

- **Execute** the process of version control (e.g., using Git) on analysis scripts, ensuring **traceability**.
- **Configure** a **message broker** (e.g., Kafka) to route data streams from an instrument.

Theory: Version Control for Traceability

Module 3.3: Execute

Git provides a full history of changes.

Your analysis script (e.g., ‘analyse.py‘) is **data**. It’s the recipe that turns raw data into a result.

Problem: ‘analyse_v1.py‘ ‘analyse_v2_final.py‘ ‘analyse_v2_final_FOR_REAL.py‘

This is a **traceability nightmare**. You no longer know which script made which figure.

Solution: Git

- ‘git commit -m ”Added new filtering step”‘
- ‘git commit -m ”Fixed bug in axis label”‘

Now you have 100% traceability.

Theory: The Message Broker

Module 3.3: Configure

A **Message Broker** (like Kafka, RabbitMQ, or MQTT) is a tool that routes data streams.

- **Instrument** → "Publishes" data to a topic (e.g., 'lab/microscope/raw_data')
- **Broker** → Receives the data.
- **ELN** → "Subscribes" to the topic and automatically ingests the data.
- **Archive** → *Also* subscribes and makes a backup.

This is the gold standard for automated, high-integrity data capture.

Interactive: Hands-On Traceability

Module 3.3: Practice

Workshop (20 min)

This is a hands-on session.

Track 1: Version Control (Git)

Goal: Execute version control.

- Create a folder.
- Create ‘script.py‘.
- Run: ‘git init‘
- Run: ‘git add .‘
- Run: ‘git commit -m ”Initial commit”‘

You just ensured traceability!

Learning Objectives (3.4)

- **Differentiate** between data annotation methods for *qualitative* vs. *quantitative* data.
- **Troubleshoot** a data pipeline error by **analysing** message broker logs.

Theory: Data Annotation Methods

Module 3.4: Differentiate

Annotation (metadata) is not one-size-fits-all.

Quantitative Data (Numbers)

Data: ‘10.5‘ **Annotation Needs:**

- **Vocabularies:** What is it? (e.g., ‘PATO:0000125‘ - ”temperature” see OLS)
- **Units:** What unit? (e.g., ‘UO:0000027‘ - ”degree Celsius”)
- **Precision:** ‘+/- 0.05‘

Goal: Consistency, machine-readability.

Qualitative Data (Text/Image)

Data: ”The sample showed significant degradation...” **Annotation Needs:**

- **Coding Schemas:** What does ”significant degradation” mean? (e.g., ‘Degradation_Level_3‘)
- **Context:** Who wrote this? When?
- **Tags:** ‘#failed_experiment‘, ‘#contamination‘

Goal: Context, findability, interpretation

Theory: Troubleshooting the Pipeline

Module 3.4: Troubleshoot

Problem: The ELN is recording gibberish, but the instrument seems fine. Where is the error?

- **Step 1:** Check the Message Broker logs. Is the data arriving at the broker correctly?
- **Yes:** The data is fine. The problem is the **ELN subscriber** (Track 2). It's misinterpreting the message.
- **No:** The data is not in the logs, or it's already gibberish. The problem is the **Instrument publisher** (Track 1). It's sending bad data.

The broker's log is your central point for **troubleshooting**.

Interactive: The Pipeline is Broken!

Module 3.4: Practice

Scenario (15 min)

Problem: Your ELN is recording temperatures as '25.3' (correct), but pressures as 'NULL'. **Logs:** The broker log shows: '"topic": "lab/sensor/temp", "payload": "25.3"' '"topic": "lab/sensor/pressure", "payload": "101.3"' **Task: Analyse & Troubleshoot**

- Is the instrument working? (Yes)
- Is the broker working? (Yes)
- What is the problem?
- **Answer:** The ELN is subscribed to 'lab/sensor/temp' but *not* to 'lab/sensor/pressure'. It's a configuration error in the subscriber.

Goal: Troubleshoot a pipeline error.

Learning Objectives (3.5)

- **Assess** the current team workflow against the DMP.
- **Compare** and **evaluate** manual vs. automated data capture, justifying the superior method based on **verifiability**.

Theory: Assessing Workflow vs. DMP

Module 3.5: Assess

Your DMP is your **plan**. Your workflow is your **reality**. You must **assess** reality against the plan.

The "DMP Audit"

Your DMP says: "All data will be annotated with a standard vocabulary."
Your Workflow is: A student is typing free-text notes into the ELN.
Result: Non-compliant. You have a bottleneck.
Solution: You must **assess** this gap and fix it (e.g., build a drop-down list in the ELN).

Theory: Manual vs. Automated (Verifiability)

Module 3.5: Compare

Verifiability (Data Integrity) is the ability to prove your data is correct.

Method 1: Manual Entry

- Student reads "10.5" from screen.
- Student types "10.5" into Excel.
- **Verifiability:** Low.
- **Error Points:** Typo? Did they misread? Did they skip a line? You can *never* be 100% sure.

Method 2: Automated Pipeline

- Instrument sends "10.5".
- Broker logs "10.5".
- ELN records "10.5".
- **Verifiability:** High.
- **Error Points:** None (in transcription). The log *proves* what the instrument sent.

Justification: The automated pipeline is superior not because it's faster, but because it is verifiable.

Interactive: Debate - Manual vs. Automated

Module 3.5: Practice

Group Evaluation (10 min)

Scenario: A senior professor says, "These new automated pipelines are a black box. I trust my PhD students to write down the numbers correctly. It's how I've always done it." **Task:**

- **Compare** the two methods.
- Form a polite, respectful argument. **Justify** why the automated pipeline is superior, using the concept of **verifiability**.

Goal: **Evaluate** two methods.

Learning Objectives (3.6)

- **Develop** a comprehensive, step-by-step **workflow diagram** that integrates automated metadata creation.
- **Design** a resilient, end-to-end RDM workflow for data collection.

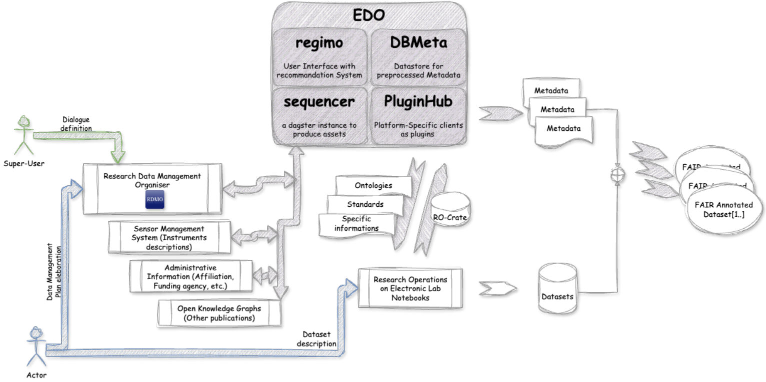
Theory: The End-to-End Workflow

- **Plan:** DMP (from RDMO)
- **Acquire:** Instrument + Instrument Registry
- **Transport:** Message Broker
- **Record:** ELN (with required fields)
- **Process:** Analysis Scripts (versioned in Git)
- **Publish:** Repository (KITOpen / OEP)

Theory: The End-to-End Workflow

Module 3.6: Develop

Let's put all the pieces from today together.



A conceptual diagram of a full RDM integrated workflow

Interactive: Design Your Workflow

Module 3.6: Practice

Activity: Whiteboard Session (20 min)

Task: As a group, **design** your ideal, resilient, end-to-end RDM workflow.

Instructions:

- Draw it on a whiteboard or paper.
- Start with the **Instrument**.
- End with the **Publication** (on KITOpen/OEP).
- **Must Include:** Your ELN, a Message Broker, and Git.
- Show where metadata is created and where it flows.

Goal: **Develop** and **Design** a comprehensive workflow diagram.

Module 3: Summary

- RDM is not just planning; it's **daily operation**.
- **ELNs** are the core tool for capturing metadata at acquisition.
- **Git** ensures traceability (integrity) for your scripts.
- **Message Brokers** provide a verifiable, automated path from instrument to ELN.
- We must **assess** our real workflow against our DMP.

End of Day 1

Outline

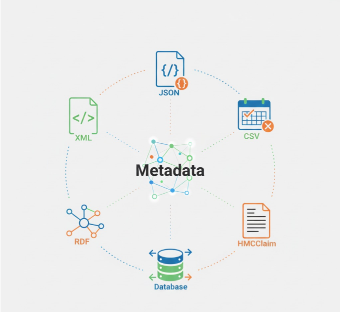
[0]

Day 2 - Part 4: Metadata Serialisation

10:00 - 10:45

Module 4 Goal

Welcome to Day 2! We will focus on the *structure* and *format* of metadata. It's not just *what* you record, but *how* you encode it.



Learning Objectives (4.1)

- **Recall** the core components of a minimal metadata standard (e.g., Dublin Core).
- **Recall** the structural differences between common metadata serialisation formats (XML, JSON, CSV).

Theory: Minimal Standard (Dublin Core)

Module 4.1: Recall

Dublin Core (DC) is a "lowest common denominator" metadata standard. It provides 15 simple fields to describe *anything*.

The 15 Dublin Core Elements

- Title
- **Creator**
- Subject
- **Description**
- Publisher
- Contributor
- **Date**
- **Type**
- **Format**
- **Identifier**
- Source
- Language
- Relation
- Coverage
- **Rights** (License)

This is the minimum you should provide for 'F' and 'R'.

Theory: Serialisation Formats

Module 4.1: Recall

How do we write this metadata down?

CSV

```
Title,Creator
My Data,A. Researcher
```

- Simple
- No hierarchy

XML

```
<Title>My Data</Title>
<Creator>A. Researcher</Creator>
```

- Verbose
- Strict (Schemas!)

JSON

```
{
  "Title": "My Data",
  "Creator": "A. Researcher"
}
```

- Web-friendly
- Less strict

Interactive: Format Brainstorm

Module 4.1: Practice

Group Brainstorm (10 min)

Task: Let's **recall** the pros and cons of these formats.

- When would **CSV** be a *good* choice for metadata?
- When would **XML** be a *better* choice than JSON? (Hint: Think about strict rules).
- Why has **JSON** become the most popular format for web APIs?

Learning Objectives (4.2)

- **Differentiate** between **descriptive** and **technical** metadata.
- **Differentiate** between the role of a **metadata schema** (structure) and a **serialisation format** (encoding).

Theory: Types of Metadata

Module 4.2: Differentiate

Not all metadata is the same.

Descriptive Metadata

What the data is.

- Title
- Creator
- Abstract
- Keywords

Purpose: Findability, Citation. ('F', 'R')

Technical Metadata

How the data was made.

- 'Software: Python 3.9'
- 'Instrument: HMC-Microscope-047'
- 'File Format: .csv'
- 'Checksum: md5:ab...c1'

Purpose: Reusability, Validation. ('I', 'R')

Theory: Schema vs. Serialisation

Module 4.2: Differentiate

This is the most important concept in this module.

- **Schema (The Blueprint):** The *rules*.
- "You MUST have a Title. The Date MUST be in 'YYYY-MM-DD' format."
- (e.g., Dublin Core, XSD, JSON-Schema)
- **Serialisation (The Building Material):** The *format* you write it in.
- "We will write these rules in XML."
- (e.g., XML, JSON, Turtle)

You can use the **Dublin Core schema** and serialise it as **XML**, **JSON**, or even **CSV**.

Interactive: Schema or Format?

Module 4.2: Practice

Quiz (10 min)

I'll name a term. You tell me: **Schema (Rule)** or **Serialisation (Format)**?

- JSON? → **Format**
- Dublin Core? → **Schema**
- XML? → **Format**
- ISO 19115 (for Geospatial data)? → **Schema**
- 'metadata.csv'? → **Format**
- JSON-Schema? → **Schema** (A schema for the JSON format!)

Learning Objectives (4.3)

- **Annotate** a raw dataset file by applying a selected discipline-specific metadata standard.
- **Annotate** a raw dataset file by applying a minimal metadata schema (Dublin Core) and exporting the result in a serialisation format (JSON).

Theory: What is Annotation?

Module 4.3: Annotate

Annotation is the act of *linking* a piece of data to its metadata.

Example Annotation

Raw Data: ‘my_experiment.csv‘ **Annotation (in a separate JSON file):**

This file now describes the raw data. This is a **Metadata Record**.

Interactive: Let's Annotate!

Module 4.3: Practice

Workshop: Annotate a File (20 min)

Task: Everyone take a simple file on your computer (a script, a CSV, a document).

1. Open a plain text editor (e.g., Notepad, VS Code).
2. Create a new file called 'metadata.json'.
3. **Annotate** your file using **Dublin Core** (Title, Creator, Date, Format).
4. Write this annotation in **JSON format** (like the previous slide).

Bonus: Add one **discipline-specific** field (e.g., 'my_field:sample_type').
Goal: **Annotate** a file and **export** to JSON.

Learning Objectives (4.4)

- **Compare** two different metadata files (XML vs. JSON) and **analyse** their efficiencies (machine readability, schema validation).
- **Compare** two different serialisation formats (XML vs. JSON) and **analyse** their efficiencies (parsing overhead).

Theory: Comparing XML vs. JSON

Module 4.4: Compare

XML (eXtensible Markup Language)

- **Pro:** Very strict validation using XSD (XML Schema).
- **Pro:** Supports Namespaces (e.g., 'dc:').
- **Con:** "Verbose" (high parsing overhead).

JSON (JavaScript Object Notation)

- **Pro:** Lightweight (low parsing overhead).
- **Pro:** Native to web browsers.
- **Con:** Validation (JSON-Schema) is less mature than XSD.

Concept: Readability vs. Validation

Module 4.4: Analyse

- **Machine Readability / Parsing Overhead:**
- JSON is "lighter" and faster for a web browser to parse. This is why it's used for web APIs.
- **Schema Validation:**
- XML is "stricter." It's often preferred by archives and large institutions (like libraries) because an XSD schema can enforce very complex rules, guaranteeing metadata integrity.

Which is "better"?

Neither. They are different tools.

- Use **JSON** for fast, lightweight data exchange (e.g., web pipelines).
- Use **XML** for strict, archival-grade metadata validation.

Interactive: Which is "Better"?

Module 4.4: Practice

Discussion (10 min)

Let's **compare** and **analyse**.

- **Scenario 1:** You are building a web dashboard to show live sensor data. Which format do you use?
- *Answer:* JSON (low overhead).
- **Scenario 2:** You are submitting your final metadata to a 100-year archive (like the national library). Which format do they probably want?
- *Answer:* XML (strict validation).

Learning Objectives (4.5)

- **Justify** the selection of a specific **controlled vocabulary** to transform free-text into standardised terms.
- **Justify** the selection of a specific **serialisation format** for a project based on pipeline constraints.

Theory: The Problem with Free-Text

Module 4.5: Justify

A "Free-Text" Metadata Record

- 'Location: Karlsruhe'
- 'Location: KIT Campus North'
- 'Location: 76131, Germany'

A machine **cannot** understand that these are the same place. The data is not Interoperable ('I').
Solution: Transform free-text using a **Controlled Vocabulary (CV)**.

A "Standardised" Metadata Record

- 'Location: https://geonames.org/2892794' (URI for Karlsruhe)

Now, a machine knows *exactly* where you mean.

Interactive: The Pipeline Choice

Module 4.5: Practice

Scenario (15 min)

Situation: Your data pipeline must feed two systems:

1. A web dashboard (needs to be fast).
2. A long-term HMC archive (needs to be strict).

Task:

- **Justify** the **serialisation format** you would choose for #1. (JSON)
- **Justify** the **serialisation format** you would choose for #2. (XML)
- How do you serve both? (Answer: You don't pick one. You *transform* your internal data *into* both formats!)

Learning Objectives (4.6)

- **Transform** a project's internal, proprietary metadata scheme into an **interoperable** format (e.g., XSLT).
- **Transform** a project's internal, proprietary metadata scheme into a **standardized serialization format**.

Theory: Metadata Transformation

Module 4.6: Transform

You will almost *never* work in the final format. You work in what's easy (your ELN, a CSV), and then **transform** it for publication.

Common Transformation Tools

- **XSLT:** A language specifically for transforming XML → XML or XML → HTML.
- **Python/R Script:** The most common. Read a CSV/JSON, use a library to write out a valid XML file.
- **LinkML:** (See next module) Can auto-generate transformers.

Interactive: Transformation Needs

Module 4.6: Practice

Discussion (10 min)

Task: Let's think about your own research.

- What is your **internal, proprietary** format? (e.g., "The '.dat' file from my instrument," "My personal Excel template.")
- What is the **standardised format** your field or repository wants? (e.g., "The OEP wants standards-compliant metadata.")
- What is the "transformation" gap you need to bridge?

Goal: Transform (conceptually) your internal format to a standard one.

Module 4: Summary

- **Metadata** has many types (Descriptive, Technical).
- **Schema** (the rules) and **Serialisation** (the format) are different.
- **XML** is for validation; **JSON** is for web-friendliness.
- We must **transform** our internal, free-text metadata into **standardised, vocabulary-controlled** records for publication.

End of Part 4

Outline

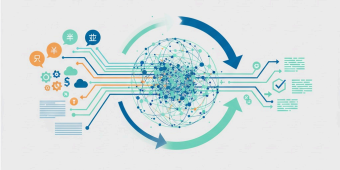
[0]

Day 2 - Part 5: Semantics & Vocabularies

10:50 - 11:35

Module 5 Goal

It's not enough for machines to *read* data. They must *understand* it. This is **Semantics**, the core of Interoperability ('I').



Semantics provides shared meaning, like a universal translator.

Learning Objectives (5.1)

- **List** three examples of knowledge organisation systems (KOS) used in RDM (controlled vocabularies, thesauri, ontologies).
- **Recall** the core components of the **Protégé** UI for defining classes and properties.

Theory: The Knowledge "Ladder" (KOS)

‚Module 5.1: List

- **1. Controlled Vocabulary (List):**
- A simple, flat list of allowed terms. (e.g., '['male', 'female', 'other']').
- **2. Thesaurus (Hierarchy):**
- A list with relationships. (e.g., 'Karlsruhe' → Broader Term: 'Baden-Württemberg').
- **3. Ontology (Logical Model):**
- A formal model of a domain. (e.g., 'Karlsruhe' *is_a* 'City'. A 'City' *has_population* 'Integer'. A 'City' *is_located_in* 'State'.)

Theory: Intro to Protégé

Module 5.1: Recall

Protégé is a free, open-source **ontology editor**. It is the standard tool for building and managing formal ontologies.

Core Components of Protégé

- **Classes:** The "things" or "concepts" in your domain. (e.g., 'Experiment', 'Sample', 'Instrument').
- **Object Properties:** Relationships between classes. (e.g., 'Experiment' → *uses_instrument* → 'Instrument').
- **Data Properties:** Relationships from a class to a value. (e.g., 'Sample' → *has_weight* → 'float').
- **Axioms:** The logical rules. (e.g., 'Experiment' *must_have_at_least_one* 'Sample').

Interactive: Demo Protégé

Module 5.1: Practice

Live Demo (10 min)

Task: Let's open Protégé and look at a simple ontology.

- **Recall:** Where is the 'Classes' tab?
- **Recall:** Where is the 'Object Properties' tab?
- We will create one class, ‘HMC_Researcher‘, and one data property, ‘has_ORCID‘.

Goal: **Recall** the core components of the UI.

Learning Objectives (5.2)

- **Explain** how a **controlled vocabulary** (CV) improves the **Interoperability** ('I') of a dataset.
- **Explain** how **LinkML** facilitates the **Interoperability** of two distinct data models.

Theory: How CVs Enable Interoperability

Module 5.2: Explain

Without a CV (Low 'I')

- ‘Researcher 1: ”temp”‘
- ‘Researcher 2: ”T”‘
- ‘Researcher 3: ”temperature”‘

A computer search for ”temperature” will **miss** the data from Researchers 1 and 2. The data is not interoperable.

With a CV (High 'I')

CV: ‘MyLab:Temperature‘ (URI)

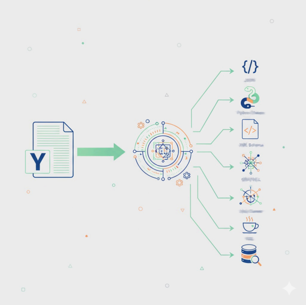
- ‘Researcher 1: MyLab:Temperature‘
- ‘Researcher 2: MyLab:Temperature‘
- ‘Researcher 3: MyLab:Temperature‘

A computer can now search for ‘MyLab:Temperature‘ and find **all three** datasets. They can be automatically combined.

Theory: Intro to LinkML

Module 5.2: Explain

LinkML is a modelling language (written in simple YAML) that helps you **design** and **validate** data models.



One Model → Many Formats

You write one simple 'schema.yaml' file... LinkML **auto-generates** all of these:

- **JSON-Schema** (for validation)
- **Python Classes** (for your scripts)
- **XML Schema (XSD)** (for your archive)
- **GraphQL** (for your web API)

It is the **ultimate** tool for Interoperability ('I').

Interactive: The Power of CVs and LinkML

Module 5.2: Practice

Discussion (10 min)

Task:

- **Explain** in your own words how a CV stops researchers from "making up" metadata terms.
- **Explain** how LinkML solves the "XML vs. JSON" problem we had in Module 4. (Answer: It generates *both* from one source!)

Goal: Explain how these tools enable 'I'.

Learning Objectives (5.3)

- **Map** free-text keywords to standardised terms in a *domain-specific thesaurus*.
- **Map** classes and properties between two existing ontologies using **Protégé**.

Theory: Mapping Free-Text

Module 5.3: Map

Mapping is the process of *translating* your messy, free-text "internal" terms to the clean, public "standard" terms.

Example: Mapping to a Thesaurus

- **Your Term:** "Pressure"
- **Thesaurus:** 'Gemet'
- **Search "Pressure":** Find 'http://.../atmospheric_pressure'
- **Mapping:** Your 'pressure' → 'owl:equivalentClass' → 'gemet:atmospheric_pressure'

You have now made your data interoperable with everyone else who uses the 'Gemet' thesaurus.

Theory: Mapping in Protégé

Module 5.3: Map

Protégé is the tool for mapping two complex *ontologies*.

Scenario:

- ‘Ontology_A‘ (from Biology) has a class: ‘Human‘
- ‘Ontology_B‘ (from Medicine) has a class: ‘Patient‘

Are these the same?

Mapping Axioms in Protégé

- ‘Ontology_A:Human‘ → ‘owl:equivalentClass‘ → ‘Ontology_B:Patient‘ (This means they are identical)
- ‘Ontology_A:Human‘ → ‘rdfs:subClassOf‘ → ‘Ontology_B:Patient‘ (This means all Humans are Patients, but not all Patients are Humans - maybe some Patients are animals?)

This logic is the core of semantic interoperability

Interactive: Mapping Workshop

Module 5.3: Practice

Workshop: Two Tracks (15 min)

Track 1: Thesaurus

Goal: Map free-text.

- Go to a public thesaurus (e.g., ‘Gemet‘, ‘AGROVOC‘).
- Find a term for ”soil”.
- Find the URI.

Track 2: Protégé (Demo)

Goal: Map two classes.

- (Live Demo)
- Load two simple ontologies.
- Create an ‘owl:equivalentClass‘ axiom between two classes.

Learning Objectives (5.4)

- **Assess** a metadata record to **identify** free-text entries that should be replaced by a CV.
- **Assess** a candidate **LinkML schema** to **identify** structural conflicts.

Theory: Assessing a Metadata Record

Module 5.4: Assess

A "Bad" Metadata Record (Free-Text)

```
{
  "title": "My Test",
  "material": "Steel",
  "measurement": "Toughness"
}
```

An "Assessed & Improved" Record (CV)

```
{
  "title": "My Test",
  "material": "emmo:SS-304",
  "measurement": "omat:Charpy_Impact_Test"
}
```

Assessment:

- What is "Steel"? (Which grade? 'SS-304'? 'SS-316'?)
- What is "Toughness"? (Charpy test? Izod test?)

We **identified** the free-text and **replaced** it with URIs from a CV.

Theory: Assessing a LinkML Schema

Module 5.4: Assess

Problem: You try to merge two data models (schemas) using LinkML.

Model A

```yaml
classes:
  Sample:
    slots:
      ID:
        identifier: true
```

Model B

```yaml
classes:
  Specimen:
    slots:
      UUID:
        identifier: true
```

Assessment: You have a **structural conflict**. ‘Sample‘ and ‘Specimen‘ are probably the same thing (‘owl:equivalentClass‘), but they have different identifier names (‘ID‘ vs. ‘UUID‘). LinkML helps you **identify** and **map** these conflicts.

Interactive: Find the Free-Text

Module 5.4: Practice

Activity (10 min)

Task: Look at this "bad" metadata record. **Assess** it and **identify** all the ambiguous free-text fields that need to be replaced by a CV.

```
{
  "author": "M. Schmidt",
  "date": "Oct 10, 2024",
  "location": "CN, Bldg 301",
  "instrument": "Zeiss Scope",
  "material": "Copper",
  "result": "Good"
}
```

Answers: ‘author‘ (needs ORCID), ‘date‘ (needs ISO format), ‘location‘ (needs URI), ‘instrument‘ (needs ID), ‘material‘ (needs URI), ‘result‘ (needs vocab).

Learning Objectives (5.5)

- **Justify** the selection of a specific **ontology** over a simpler **controlled vocabulary**.
- **Justify** the decision to use an advanced feature in **Protégé** (e.g., reasoners).

Module 5.5: Justify

When do you need the complexity of an ontology?

Use a Controlled Vocabulary (CV)

When: You need to *standardise input*.

- Drop-down lists.
- Simple tags.
- Harmonising keywords.

Query: "Find all data tagged with ‘Karlsruhe‘."

Use an Ontology

When: You need to ask *complex, logical questions*.

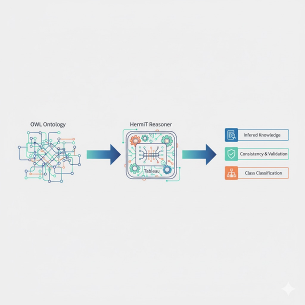
- ‘Experiment‘ *uses_material* ‘Steel‘
- ‘Steel‘ *has_component* ‘Iron‘

Query: "Find all experiments that used a material which has Iron as a component."

Justification: You *must* use an ontology if your research questions require logical inference.

Theory: Protégé Reasoners

Module 5.5: Justify



A reasoner is a "logic engine" for your ontology.

You state the facts. The reasoner finds the hidden knowledge.

- **You state:** 'Karlsruhe' *is_a* 'City'.
- **You state:** 'City' *is_a* 'Populated_Place'.
- **Reasoner infers:** 'Karlsruhe' *is_a* 'Populated_Place'. (Transitivity)

Justification: You *must* use a reasoner to **validate the semantic consistency** of your model. If you state 'A' is a 'B', and 'B' is a 'C', but also 'A' *cannot* be a 'C' (disjoint), the reasoner will raise an error and tell you your logic is broken.

Interactive: Vocab or Ontology?

Module 5.5: Practice

Scenario (15 min)

Task: For each scenario, **justify** your choice: a simple **CV** or a full **Ontology**?

- **Scenario 1:** You want to standardise the "Country" field in a user sign-up form.
- *Answer:* **CV**. (A simple list).
- **Scenario 2:** You want to model the entire energy grid, including power plants, substations, and the flow of electricity between them, to find weak points.
- *Answer:* **Ontology**. (Needs complex relationships).
- **Scenario 3:** You want to validate that your semantic mappings between two models are logical and not contradictory.
- *Answer:* **Ontology with a Reasoner**.

Learning Objectives (5.6)

- **Develop** a simple, internal **controlled vocabulary** (CV) for an ELN template.
- **Develop** a **LinkML data model** that expresses semantic similarities identified via mapping.

Interactive: Draft an Internal CV

Module 5.6: Practice

Activity (10 min)

Task: Let's **develop** a simple internal CV for our ELN. **Field:** ‘experiment_status‘
Brainstorm: What are the allowed terms?

- ‘planning‘
- ‘active‘
- ‘paused‘
- ‘completed‘
- ‘failed‘

Result: You have just created a CV! You can now build this into your ELN as a drop-down list to prevent ”free-text” like ”done” or ”finished”.

Interactive: Draft a LinkML Model

Module 5.6: Practice

Challenge: Draft a LinkML Model (15 min)

Task: Remember our mapping: ‘Model A: Sample‘ and ‘Model B: Specimen‘.
Goal: **Develop** a LinkML model that merges them.
Solution (YAML):

```yaml
classes:
  MyMergedClass:
    # We map to BOTH ontologies
    is_a: ModelA:Sample
    exact_mappings:
      - ModelB:Specimen
    slots:
      # We merge the slots
      - ID
      - UUID
```

This LinkML model now formally expresses the semantic similarity.

Module 5: Summary

- **Semantics** ('I') is about shared *meaning*, not just format.
- We use a "ladder" of tools: **CVs** (lists), **Thesauri** (hierarchy), **Ontologies** (logic).
- **Protégé** is the tool for building and mapping ontologies.
- **LinkML** is the tool for building and merging data models.
- We must **map** our free-text to standard terms.

End of Part 5

Outline

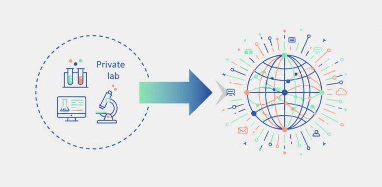
[0]

Day 2 - Part 6: Publishing Your Metadata

12:15 - 13:00

Module 6 Goal

The final step! Making your data public, findable, and accessible using our key platforms: **KITOpen** and the **Open Energy Platform (OEP)**.



From your lab (private) to the world (public).

Learning Objectives (6.1)

- **Identify** the mandatory submission fields for **KITOpen**.
- **List** the specific **domain-relevant metadata properties** unique to the **Open Energy Platform (OEP)**.

Theory: Institutional vs. Domain Repositories

Module 6.1: Identify/List

KITOpen (Institutional)

Role: Archives *all* research output from KIT (papers, data, theses).

Metadata: General-purpose (Dublin Core).

Mandatory Fields:

- Title
- Creator (with KIT affiliation)
- Publication Year
- Resource Type (e.g., "Dataset")
- License

Open Energy Platform (Domain)

Role: Archives *only* energy research data, from *all* institutions.

Metadata: Domain-specific.

Domain-Relevant Properties:

- ‘energy_carrier_type‘
- ‘technology_type‘
- ‘geographic_region (NUTS-3)‘
- ‘temporal_resolution‘

Interactive: Repository Scavenger Hunt

Module 6.1: Practice

Activity (10 min)

Task: Open the websites for ‘KITOpen‘ and ‘OEP‘.

- Find the "Upload" or "Submit" page for new data.
- **Identify** 3 mandatory fields in KITOpen.
- **List** 3 *domain-specific* fields in OEP that KITOpen does not have.

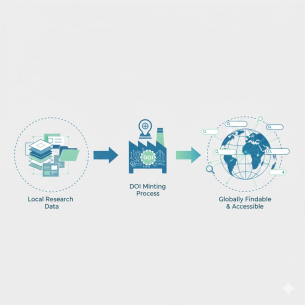
Discussion: Why does OEP ask for ‘temporal_resolution‘ while KITOpen doesn’t?

Learning Objectives (6.2)

- **Summarise** the process by which **KITOpen** assigns a PID (DOI) and fulfils 'F'.
- **Illustrate** how the community standards enforced by **OEP** enhance 'I'.

Theory: KITOpen and 'Findable'

Module 6.2: Summarise



The DOI Minting Process

1. You upload your data to KITOpen.
2. You fill in the metadata (Title, Creator, etc.).
3. You click "Publish."
4. KITOpen (as a DataCite member) sends your metadata to the DOI registration agency.
5. A new, unique **DOI** (e.g., '10.5445/IR/100...') is created and permanently linked to your data's metadata page on KITOpen.

Your data is now globally **Findable**.

Theory: OEP and 'Interoperable'

Module 6.2: Illustrate

The OEP's community standards (its metadata schema) are its power.

How OEP Enhances 'I'

Problem:

- You have a dataset of German power grids.
- A researcher from France has a dataset of French power grids.

Solution:

- The OEP *forces* both of you to use the *same metadata fields* (e.g., 'energy_carrier_type') and the *same vocabularies*.
- **Result:** A third person can now write *one script* to query and automatically combine both your dataset and the French dataset.

This is **interoperability** in action, enforced by a domain repository.

Interactive: The Role of Each Repo

Module 6.2: Practice

Discussion (10 min)

Task:

- **Summarise:** Why is getting a DOI from KITOpen better than just putting data on your lab website?
- **Illustrate:** Draw a simple diagram showing how the OEP's common standard allows two different datasets (A and B) to be combined by a third party (C).

Learning Objectives (6.3)

- **Upload** a dataset and metadata file to **KITOpen**.
- **Populate** an **OEP** metadata template using controlled vocabularies.

Interactive: Mock Submission (KITOpen)

Module 6.3: Practice

Workshop (10 min)

Task: (Conceptual or Live Demo)

1. Take the 'metadata.json' file you created in Module 4.
2. Go to the KITOpen submission form.
3. **Upload** your (dummy) data file.
4. Manually copy the fields (Title, Creator, etc.) from your JSON file into the corresponding fields in the KITOpen web form.

Goal: Upload (conceptually) a dataset and populate the metadata.

Interactive: Mock Submission (OEP)

Module 6.3: Practice

Workshop (10 min)

Task: Go to the OEP submission template. **Challenge:** Find the field for "Geographic Region."

- It likely requires a **controlled vocabulary** (e.g., NUTS codes).
- You can't just type "Karlsruhe."
- **Find the NUTS code for Karlsruhe.** (It's 'DE122').

Goal: Populate an OEP template using a required CV.

Learning Objectives (6.4)

- **Differentiate** between the institutional schema (KITOpen) and a general schema (Dublin Core).
- **Compare** the required access policy settings between KITOpen and OEP.

Theory: Schema Mapping (DC vs. KITOpen)

Module 6.4: Differentiate

- **Dublin Core (DC):** Is general. It has 15 fields.
- **KITOpen Schema:** Is specific. It *includes* all of Dublin Core (this is called "DC-compliance"), but adds **institutional** fields.
- **Analysis:**
  - 'DC:Creator' → 'KITOpen:Creator' (Direct map)
  - 'DC:Rights' → 'KITOpen:License' (Direct map)
  - 'KITOpen:KIT_Affiliation' → **No equivalent in DC!**
  - 'KITOpen:Project_ID (DFG)' → **No equivalent in DC!**

Conclusion

KITOpen's schema is an *extension* of Dublin Core, enriched with fields that matter to KIT and its funders.

Theory: Comparing Access Policies

Module 6.4: Compare

KITOpen Access Policy

- **Focus:** Institutional mandate.
- **Default:** As open as possible, as closed as necessary.
- **Options:** Open, Embargoed, Restricted (to KIT), Closed.
- Very flexible to support all disciplines (from humanities to engineering).

OEP Access Policy

- **Focus:** Community building.
- **Default:** Open Access (CC-BY).
- **Culture:** Strong push for openness. Restricted data is rare.
- Less flexible, as it serves a community with a shared goal of open, interoperable data.

Interactive: Institutional vs. Domain

Module 6.4: Practice

Debate (15 min)

Scenario: You only have time to publish your energy dataset in *one* repository.

Group 1: Argue for KITOpen

- Fulfils KIT mandate
- Long-term preservation guarantee
- Flexible access

Group 2: Argue for OEP

- Reaches your *community*
- Higher impact in your field
- Better, richer metadata

Goal: Compare policies and **Differentiate** schemas.

Learning Objectives (6.5)

- **Appraise** a draft KITOpen record to ensure the selected **reuse license** (e.g., CC-BY) balances policy and FAIR principles.
- **Justify** the decision to **cross-reference** the OEP entry with a publication, assessing the benefit to **Findability**.

Theory: Appraising a License

Module 6.5: Appraise

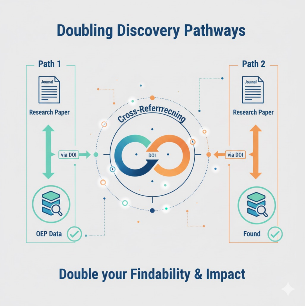
When choosing a license on KITOpen, you must **appraise** the options.

- **CC0 / CC-BY: FAIR Principles**: Excellent! (Maximises 'R'eusable), **Institutional Policy:** Excellent! (Meets open access goals).
- **CC-BY-NC-ND**: **FAIR Principles:** Poor! This license *prevents* reusability ('R') by stopping others from combining your data or using it in new tools, **Institutional Policy:** Acceptable, but not preferred.

Appraisal: Unless you have a strong legal/commercial reason, CC-BY is the best choice.

Theory: The Power of Linking

Module 6.5: Justify



Justification: Why link your OEP data to your journal paper?

- **Path 1:** Researcher finds your **Paper**. The paper links to the **OEP** data (via DOI). → **Data is Found!**
- **Path 2:** Researcher finds your **OEP Data**. The metadata links to the **Paper** (via DOI). → **Paper is Found!**

By **cross-referencing**, you double the pathways to your work, dramatically increasing its **Findability** and potential **Impact**.

Interactive: Appraise and Justify

Module 6.5: Practice

Discussion (10 min)

Task 1: Appraise

- Your colleague wants to use a 'CC-BY-ND' (No-Derivatives) license "so nobody can misrepresent my data." **Appraise** this choice. Is it good for FAIR?

Task 2: Justify

- **Justify** (in one sentence) why you must include your ORCID iD in your KITOpen submission. (Hint: It links your *data* to your *person*).

Learning Objectives (6.6)

- **Draft** the final, publicly visible **data citation** and **abstract** for a dataset, optimising for findability.
- **Synthesize** a final publication strategy that integrates **KITOpen** and **OEP**.

Theory: Writing a "Data Abstract"

Module 6.6: Draft

Your Data Abstract is **not** your Paper Abstract. It's "SEO" for your data.

Bad Data Abstract

"This file contains the data for our recent paper. See the paper for details."

Problem: No keywords. No context. Not findable.

Good Data Abstract

"This dataset contains raw data from a 2025 simulation of the German power grid, focusing on the NUTS-3 region of Karlsruhe (DE122). Data includes hourly load profiles and wind generation, modelled using [Software]... "

Result: Rich with keywords. Highly findable.

Theory: The "Both" Strategy

Module 6.6: Synthesise

You don't have to choose between KITOpen and OEP. The best strategy is to use **both**.

The "HMC/KIT" Best Practice Strategy

1. **Archive (Integrity):** Deposit the full, final, static dataset on **KITOpen**. This fulfils your institutional duty and gets a permanent DOI for long-term archiving.
2. **Index (Interoperability):** Create a *new metadata record* on **OEP**.
3. **Link (Findability):** In the OEP record, do **not** re-upload the data. Instead, point to the **KITOpen DOI**.

Result: You get the long-term integrity of KITOpen *and* the domain-specific findability of OEP.

Interactive: Your Final Strategy

Module 6.6: Practice

Final Activity (15 min)

Task 1: Draft

- Take 5 minutes and **draft** a "good" data abstract (2-3 sentences) for your own research data. Focus on keywords.

Task 2: Synthesise

- **Synthesize** your final publication strategy. How will you use both your institutional and domain repositories?
- Share your plan.

Workshop Summary

Day 1: Foundations

- Integrity → FAIR
- DMP → RDMO
- Operations → ELN, Git, Pipelines

Day 2: Metadata

- Serialisation (XML, JSON)
- Semantics (CV, Ontology)
- Publishing (KITOpen, OEP)

All Goals Achieved!

You can now connect Data Integrity to FAIR, build a DMP in RDMO, integrate RDM into operations, serialise and semantically enrich metadata, and publish on KITOpen and OEP.

Thank You

Questions?

Helmholtz Metadata Collaboration (HMC)
Karlsruhe Institute of Technology (KIT)